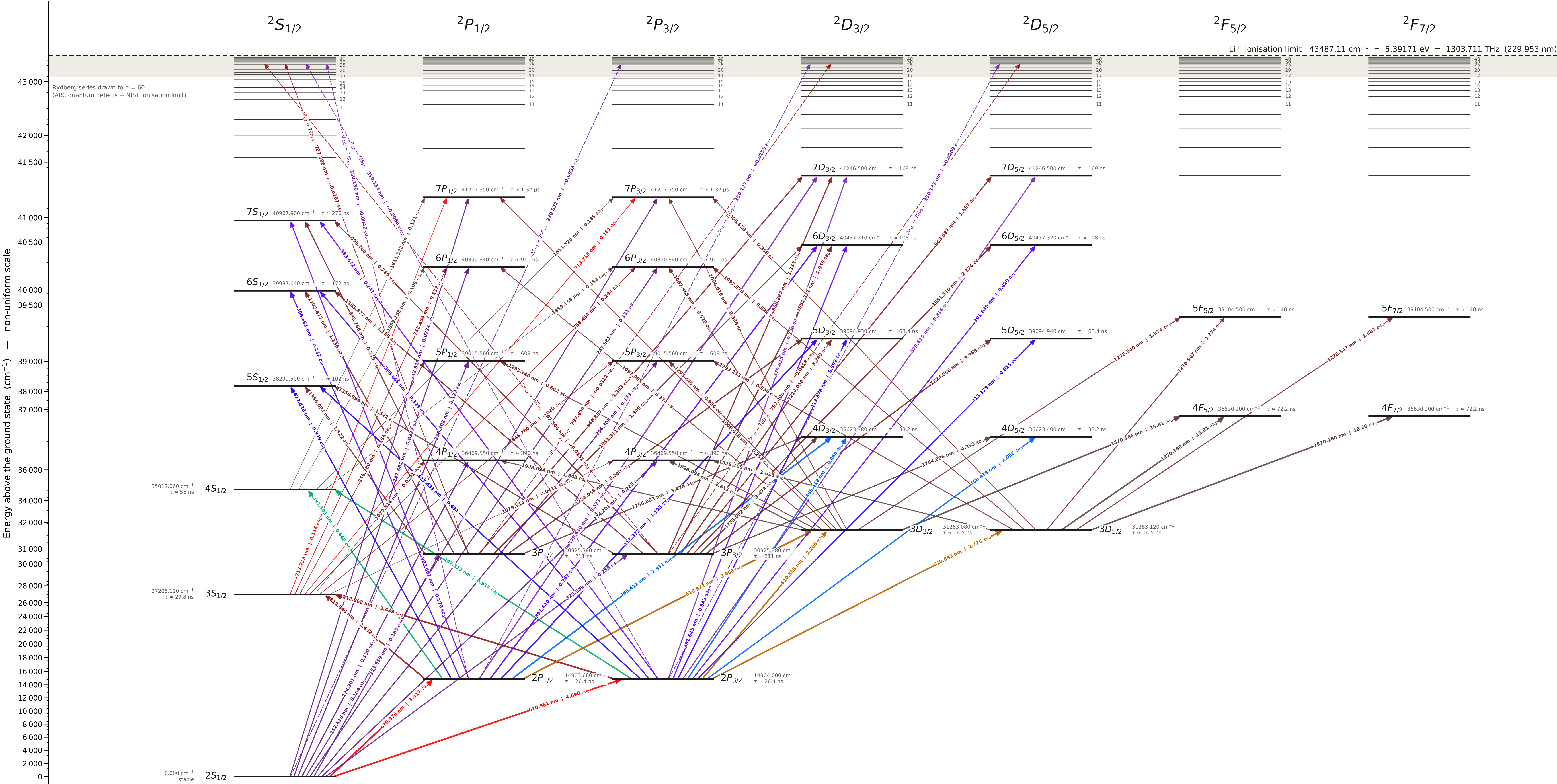
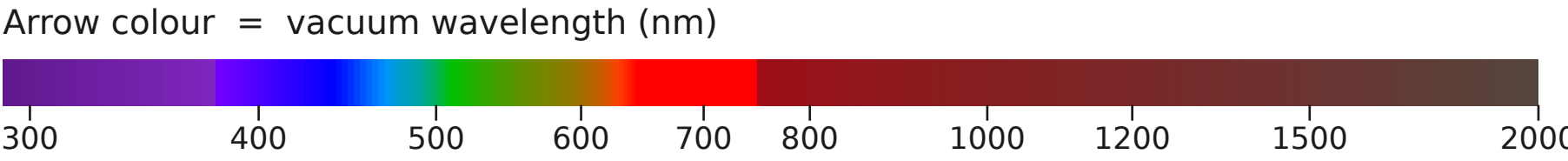


# Lithium-6

Arrow colour = vacuum wavelength (nm)  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked  $\approx$ ).



## Key transitions

| Transition                      | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $\tilde{\nu}$ (cm⁻¹) | $ \langle J  er  J' \rangle $ (ea <sub>0</sub> ) | A (s⁻¹)   | $\Gamma/2\pi$ up. (MHz) | branch (%) | Typical use |
|---------------------------------|--------------------|--------------------|-------------|----------------------|--------------------------------------------------|-----------|-------------------------|------------|-------------|
| $2P_{1/2} \rightarrow 3D_{3/2}$ | 610.5222           | 610.3533           | 491.0427    | 16379.420            | 5.066                                            | 5.714e+07 | 10.960                  | 83.3       |             |
| $2S_{1/2} \rightarrow 2P_{3/2}$ | 670.9608           | 670.7756           | 446.8107    | 14904.000            | 4.690                                            | 3.689e+07 | 6.024                   | 100.0      |             |
| $2S_{1/2} \rightarrow 2P_{1/2}$ | 670.9761           | 670.7909           | 446.8005    | 14903.660            | 3.317                                            | 3.689e+07 | 6.024                   | 100.0      |             |
| $2P_{3/2} \rightarrow 3S_{1/2}$ | 812.8680           | 812.6446           | 368.8083    | 12302.120            | 3.439                                            | 2.231e+07 | 5.344                   | 66.7       |             |
| $2P_{1/2} \rightarrow 4D_{3/2}$ | 460.4111           | 460.2821           | 651.1408    | 21719.720            | 1.931                                            | 1.935e+07 | 4.791                   | 64.5       |             |
| $2P_{3/2} \rightarrow 3D_{5/2}$ | 610.5334           | 610.3644           | 491.0337    | 16379.120            | 2.775                                            | 1.143e+07 | 10.960                  | 100.0      |             |
| $2P_{3/2} \rightarrow 3D_{3/2}$ | 610.5349           | 610.3659           | 491.0325    | 16379.080            | 2.266                                            | 1.143e+07 | 10.960                  | 16.7       |             |
| $2P_{1/2} \rightarrow 3S_{1/2}$ | 812.8456           | 812.6221           | 368.8185    | 12302.460            | 2.432                                            | 1.116e+07 | 5.344                   | 33.3       |             |
| $2P_{1/2} \rightarrow 5D_{3/2}$ | 413.3723           | 413.2557           | 725.2360    | 24191.270            | 1.123                                            | 9.040e+06 | 2.512                   | 57.6       |             |
| $2P_{3/2} \rightarrow 4S_{1/2}$ | 497.3130           | 497.1743           | 602.8245    | 20108.060            | 0.917                                            | 6.919e+06 | 2.840                   | 38.8       |             |
| $2P_{1/2} \rightarrow 6D_{3/2}$ | 391.6401           | 391.5292           | 765.4796    | 25533.650            | 0.767                                            | 4.964e+06 | 1.478                   | 54.0       |             |
| $2P_{3/2} \rightarrow 4D_{5/2}$ | 460.4179           | 460.2889           | 651.1312    | 21719.400            | 1.058                                            | 3.871e+06 | 4.791                   | 77.5       |             |
| $2P_{3/2} \rightarrow 4D_{3/2}$ | 460.4183           | 460.2893           | 651.1306    | 21719.380            | 0.864                                            | 3.871e+06 | 4.791                   | 12.9       |             |
| $2P_{1/2} \rightarrow 4S_{1/2}$ | 497.3046           | 497.1659           | 602.8347    | 20108.400            | 0.648                                            | 3.460e+06 | 2.840                   | 19.4       |             |
| $2P_{3/2} \rightarrow 5S_{1/2}$ | 427.4326           | 427.3124           | 701.3795    | 23395.500            | 0.494                                            | 3.170e+06 | 1.559                   | 32.1       |             |
| $2P_{1/2} \rightarrow 7D_{3/2}$ | 379.6098           | 379.5020           | 789.7385    | 26342.840            | 0.573                                            | 3.041e+06 | 0.941                   | 52.0       |             |
| $2P_{3/2} \rightarrow 5D_{5/2}$ | 413.3779           | 413.2613           | 725.2261    | 24190.940            | 0.615                                            | 1.810e+06 | 2.512                   | 69.1       |             |

## Rydberg excitation wavelengths (vacuum, nm)

| Lower $\rightarrow$ series      | n = 30   | n = 40   | n = 50   | n = 70   | n = 100  | n $\rightarrow \infty$ | d <sub>70</sub> ARC |
|---------------------------------|----------|----------|----------|----------|----------|------------------------|---------------------|
| $2S_{1/2} \rightarrow nP_{3/2}$ | 230.602  | 230.317  | 230.186  | 230.072  | 230.011  | 229.953                | 0.0033              |
| $2P_{3/2} \rightarrow nS_{1/2}$ | 351.397  | 350.715  | 350.404  | 350.134  | 349.992  | 349.857                | 0.0060              |
| $2P_{3/2} \rightarrow nD_{5/2}$ | 351.356  | 350.698  | 350.395  | 350.131  | 349.991  | 349.857                | 0.0209              |
| $3S_{1/2} \rightarrow nP_{3/2}$ | 618.862  | 616.817  | 615.877  | 615.060  | 614.628  | 614.213                | 0.0052              |
| $3P_{3/2} \rightarrow nS_{1/2}$ | 804.085  | 800.527  | 798.905  | 797.506  | 796.770  | 796.068                | 0.0151              |
| $3P_{3/2} \rightarrow nD_{3/2}$ | 803.872  | 800.439  | 798.860  | 797.490  | 796.764  | 796.068                | 0.0418              |
| $3D_{3/2} \rightarrow nP_{3/2}$ | 827.699  | 824.045  | 822.367  | 820.912  | 820.142  | 819.404                | 0.0077              |
| $3D_{5/2} \rightarrow nF_{7/2}$ | 827.672  | 824.034  | 822.361  | 820.910  | 820.141  | 819.404                | 0.0449              |
| $4S_{1/2} \rightarrow nP_{3/2}$ | 1197.210 | 1189.582 | 1186.088 | 1183.064 | 1181.465 | 1179.933               | 0.0070              |
| $4P_{3/2} \rightarrow nS_{1/2}$ | 1450.888 | 1439.347 | 1434.111 | 1429.610 | 1427.245 | 1424.996               | 0.0276              |
| $4P_{3/2} \rightarrow nD_{5/2}$ | 1450.194 | 1439.061 | 1433.965 | 1429.558 | 1427.228 | 1424.996               | 0.0670              |
| $4D_{5/2} \rightarrow nP_{3/2}$ | 1483.369 | 1471.676 | 1466.331 | 1461.712 | 1459.272 | 1456.937               | 0.0184              |
| $4D_{3/2} \rightarrow nF_{7/2}$ | 1483.284 | 1471.641 | 1466.314 | 1461.706 | 1459.270 | 1456.937               | 0.0837              |
| $4F_{7/2} \rightarrow nD_{5/2}$ | 1484.786 | 1473.117 | 1467.778 | 1463.161 | 1460.720 | 1458.382               | 0.0080              |
| $5S_{1/2} \rightarrow nP_{3/2}$ | 1974.212 | 1953.555 | 1944.148 | 1936.037 | 1931.758 | 1927.668               | 0.0086              |

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. d<sub>70</sub> =  $|\langle J||er||J' \rangle|$  to n = 70 in ea<sub>0</sub> (calculated).  
S  $\rightarrow$  nP values come from a near-cancelling integral: treat them as an order of magnitude.

## <sup>6</sup>Li hyperfine structure (I = 1)

| Level      | A (MHz) | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|------------|---------|---------|--------------------------------------------------------|
| $2S_{1/2}$ | 152.137 | -       | F=1/2: -152.14 F=3/2: +76.07                           |
| $2P_{1/2}$ | 17.375  | -       | F=1/2: -17.38 F=3/2: +8.69                             |
| $2P_{3/2}$ | -1.155  | -0.100  | F=1/2: +2.76 F=3/2: +1.25 F=5/2: -1.76                 |
| $3P_{3/2}$ | -0.400  | -       | F=1/2: +1.00 F=3/2: +0.40 F=5/2: -0.60                 |

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).

$\lambda$  and  $\nu$  from NIST level energies (centre of gravity, no hyperfine structure);  $\lambda$  air uses standard air (Ciddor).  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ . Branch: share of the upper level's decays that takes this line.